

RESEARCH

RESEARCH QUESTION: Effect of homebased exercise on fatigue levels in breast cancer patient on chemotherapy

PICO framework

P: Early-stage breast cancer patient (age 18+) on chemotherapy and medically cleared for physical activity

I: structured Home-based walking programme 3 sessions a week for 12 weeks, 30 minutes/session

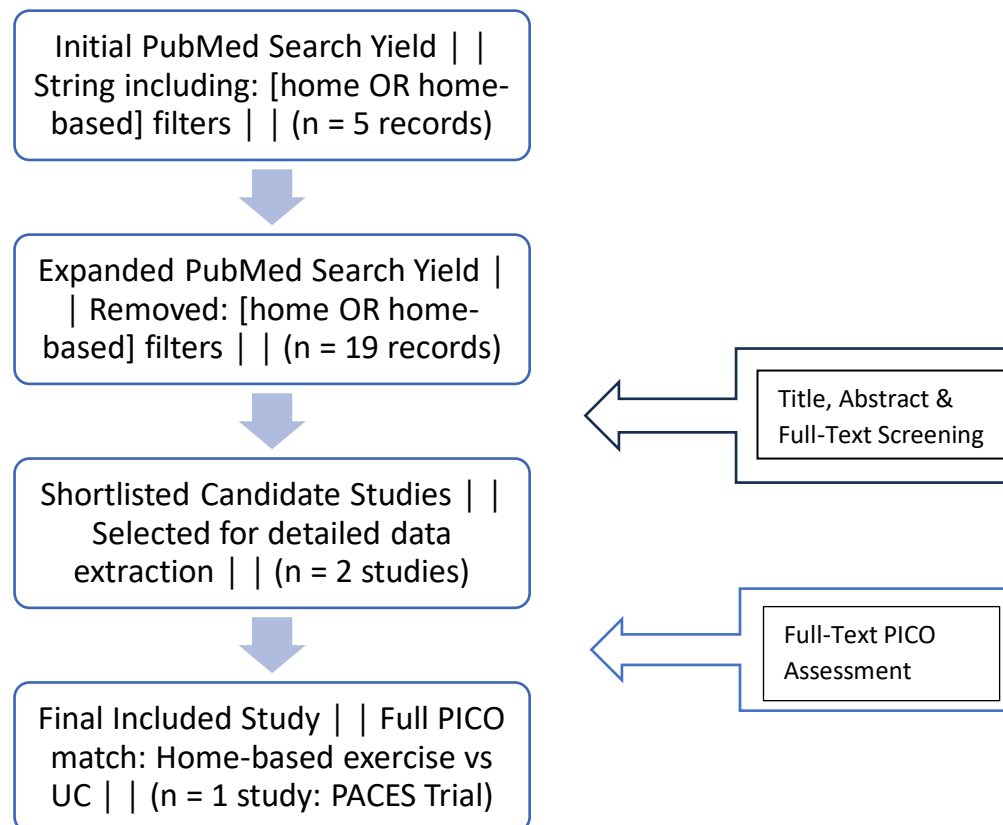
C: Usual care (no structured exercise intervention)

O: Cancer related fatigue measured via validated fatigue scale

Boolean string

("Breast Neoplasms"[MeSH] OR "Breast Neoplasm"[tiab]) AND "Chemotherapy, Adjuvant"[Mesh] AND "Fatigue"[Mesh] AND "Exercise"[Mesh]

PUBMED SEARCH



STUDIES

1. Effect of Low-Intensity Physical Activity and Moderate- to High-Intensity Physical Exercise During Adjuvant Chemotherapy on Physical Fitness, Fatigue, and Chemotherapy Completion Rates: Results of the PACES Randomized Clinical Trial
<https://pubmed.ncbi.nlm.nih.gov/25918291/>

STUDY 1

Paper title: Effect of Low-Intensity Physical Activity and Moderate- to High-Intensity Physical Exercise During Adjuvant Chemotherapy on Physical Fitness, Fatigue, and Chemotherapy Completion Rates: Results of the PACES Randomized Clinical Trial. (Section: Title)

Sample size (total, and per arm if reported):

- **Total:** n=230
- **OnTrack (Moderate-to-high intensity):** n=76
- **Onco-Move (Low-intensity):** n=77
- **Usual care (UC):** n=77
- (Section: "Abstract", "Results - Patients", "Table 2")

Tumor stage(s) included:

- Stage I: 12 (6%).
- Stage II: 109 (47%).
- Stage III: 109 (47%).
- (Section: "Results - Participants", "Table 2")

Chemotherapy regimen (name/protocol as reported):

- Adjuvant chemotherapy combinations including anthracyclines, taxanes, alkylating agents, and antimetabolites. (Section: "Results - Chemotherapy and Trastuzumab Completion Rates")

Exercise intervention protocol:

- **Frequency (sessions/week):**
 - **Onco-Move:** 5 days per week.
 - **OnTrack:** Two supervised sessions per week; participants were also encouraged to be active independently for 30 minutes on 5 days each week.
- **Intensity (as reported, e.g. %HRmax, RPE, or descriptive):**
 - **Onco-Move:** Low-intensity; 12 to 14 on the Borg Scale of perceived exertion.
 - **OnTrack:** Moderate-to-high intensity; resistance training at 80% of one repetition maximum and aerobic training at 50% to 80% of maximal workload; intensity adjusted using the Borg Scale (threshold <12 for increase and >16 for decrease).
- **Type (e.g. walking, resistance, aerobic):**

- **Onco-Move:** Home-based physical activity (walking program).
- **OnTrack:** Combined resistance (six large muscle groups) and aerobic exercise.
- **Time (minutes/session, total duration in weeks):**
 - **Onco-Move:** At least 30 minutes per day.
 - **OnTrack:** 50 minutes per supervised session (20 minutes resistance and 30 minutes aerobic).
 - **Total duration:** Both interventions started with the first cycle of chemotherapy and continued until 3 weeks after the last cycle. The mean prescribed length of chemotherapy ranged from 116.7 to 119.9 days across groups.
- (Section: "Patients and Methods - Interventions", "Table 6")

Drop-out rate (number/% and reasons if given):

- **At completion of chemotherapy (T1):** 26 patients (11.3%) did not have outcome data.
- **By group at T1:** OnTrack = 5; Onco-Move = 8; Usual Care = 11.
- **Reasons (T1):** Felt too ill (n=5), physical accident unrelated to trial (n=2), physical accident related to trial (n=1), unwilling (n=14), neuropathy (n=1), emigrated (n=1), and unknown (n=2).
- (Section: "Results - Patients", "Figure 1")

Primary outcome measure (name of validated scale used):

- **Cardiorespiratory fitness:** Steep Ramp Test (maximal short exercise capacity) and an endurance test.
- **Muscle strength:** microFET handheld dynamometer (elbow flexion and knee extension), JAMAR grip strength dynamometer, and the 30-second chair stand test.
- **Fatigue:** Multidimensional Fatigue Inventory (MFI) and the Fatigue Quality List (FQL).
- (Section: "Patients and Methods - Timing of Assessments and Study Measures", "Table 1")

Primary outcome result (direction of effect, and p-value or CI if reported):

- **Cardiorespiratory fitness:** At T1, OnTrack had significantly higher maximal short exercise capacity than UC (Adjusted Mean Difference [AMD] 22.1; 95% CI, 8.5 to 35.6; P=.001 and Onco-Move (AMD 15.4; 95% CI, 3.0 to 27.7; P=.015).
- **Fatigue:** At T1, OnTrack reported significantly less physical fatigue than UC (AMD -2.7; 95% CI, -4.0 to -1.4; P< .001) and Onco-Move (AMD -1.6; 95% CI, -2.9 to -0.2; P= .021).
- **Muscle strength:** OnTrack had significantly greater strength than UC and Onco-Move at T1 across measures, such as elbow flexion (vs UC AMD 7.0; 95% CI, 2.6 to 11.3; P= .002).
- **Overall:** Physical fitness levels were generally maintained in OnTrack but declined in the UC and Onco-Move groups.
- (Section: "Results - Cardiorespiratory Fitness, Muscle Strength, and Fatigue", "Table 3")

DATA EXTRACTION

DATA	STUDY 1
STUDY	Effect of Low-Intensity Physical Activity and Moderate- to High-Intensity Physical Exercise During Adjuvant Chemotherapy on Physical Fitness, Fatigue, and Chemotherapy Completion Rates: Results of the PACES Randomized Clinical Trial
YEAR	2015
SAMPLE SIZE TOTAL+PER RELEVANT ARM	Total N = 230. Onco-Move: n = 77; UC: n = 77
TUMOR STAGE	Breast cancer. Stage I: 6%; Stage II: 47%; Stage III: 47%
CHEMO REGIMEN	Adjuvant combinations of anthracyclines, taxanes, alkylating agents, and antimetabolites
EXERCISE TYPE/ MODALITY(RELEVANT ARM ONLY)	Onco-Move: Home-based, low-intensity, individualized self-managed physical activity program
FITT	F: 5 days/week; I: 12–14 on Borg Scale; T: Walking (physical activity); T: At least 30 min/session for duration of chemo
COMPARATOR	Usual Care (UC): Varied by hospital, but did not involve routine exercise
FATIGUE SCALE USED	Multidimensional Fatigue Inventory (MFI); Fatigue Quality List (FQL); EORTC QLQ-C30
FATIGUE RESULT(RELEVANT COMPARISON ONLY WITH P VALUE)	Onco-Move vs. UC (at T1): MFI Physical Fatigue: P = .10 (AMD -1.1); EORTC QLQ-C30 Fatigue: P = .003 (AMD -10.6)
DROP-OUT RATE	11.3% (26 patients) at completion of chemotherapy (T1)
NOTES/LIMITATIONS	Onco-Move showed significant benefits in physical functioning, nausea/vomiting, pain, and return to work compared to UC. "Result is inconsistent across fatigue instruments — significant on EORTC QLQ-C30 but not on MFI, so conclusion depends on which scale is prioritized."

FINDING

This search and screening process, applying strict PICO criteria, yielded only one study (PACES) that directly compares home-based exercise to usual care on fatigue outcomes — highlighting a gap in the current literature for this specific comparison.

SAMPLE MANUSCRIPT

This systematic review investigates whether a structured, home-based walking program reduces cancer-related fatigue in adult early-stage breast cancer patients during adjuvant chemotherapy. The target intervention was a 12-week walking program (30 minutes per session, three times per week) compared against usual care without structured exercise. Fatigue outcomes were measured using validated clinical scales.

To identify relevant studies, a systematic search was conducted in PubMed using a combination of Medical Subject Headings (MeSH) and title/abstract keywords ([tiab]) joined by Boolean logic. The exact search string was:

("Breast Neoplasms"[MeSH] OR "Breast Neoplasm"[tiab]) AND "Chemotherapy, Adjuvant"[Mesh] AND "Fatigue"[Mesh] AND "Exercise"[Mesh]

The selection process was refined in stages. An initial search using restrictive "home" or "home-based" keywords yielded only 5 records, indicating low recall that risked missing valid studies. Removing these specific terms expanded the yield to 19 records. Subsequent title, abstract, and full-text screening narrowed this pool to 2 candidate studies. After full data extraction, only one study—the PACES trial—fully met every PICO requirement through its home-based Onco-Move intervention arm. In the PACES trial, fatigue results varied across measurement tools, showing a statistically significant benefit on the EORTC QLQ-C30 scale, but not on the Multidimensional Fatigue Inventory.

This progression highlights the classic precision-recall tradeoff in literature searching: overly narrow terms increase precision but sacrifice yield. Ultimately, this review reveals a significant literature gap, showing a surprising scarcity of randomized controlled trials specifically evaluating structured home-based walking programs for patients undergoing active adjuvant chemotherapy.